

Signet 2551 Magmeter Flow Sensor



Available in a variety of wetted materials and ideal for pipe sizes up to DN900 (36 in.)



The Signet 2551 Magmeter is an insertion style magnetic flow sensor that features no moving parts. The patented* sensor design is available in corrosion-resistant materials to provide long-term reliability with minimal maintenance costs. Material options include PP with stainless steel, PVDF with Hastelloy-C, or PVDF with Titanium. Utilizing the comprehensive line of Signet installation fittings, sensor alignment and insertion depth is automatic. These versatile, simple-to-install sensors deliver accurate flow measurement over a wide dynamic range in pipe sizes ranging from DN15 to DN900 (½ to 36 inches), satisfying the requirements of many diverse applications.

Signet 2551 Magmeters offer many output options of frequency/digital (S³L) or 4 to 20 mA which are available on both the blind and display versions. The frequency or digital (S³L) sensor output can be used with Signet's extensive line of flow instruments while the 4 to 20 mA output can be used for a direct input to PLCs, chart recorders, etc. Both the 4 to 20 mA output and digital (S³L) sensor interface is available for long distance signal transmission. An additional benefit is the empty pipe detection which features a zero flow output when the sensors are not completely wetted. Also, the frequency output is bi-directional while the 4 to 20 mA output can be set for uni- or bi-directional flow using the display or the 3-0250 USB to Digital (S³L) Configuration/Diagnostic setup tool which connects to PCs for programming capabilities.

In addition the display version of the 2551 Magmeter is available with relays and features permanent and resettable totalizer values which can be stored and seen on the display. Also, the display contains multi-languages with English, Spanish, German, French, Italian and Portuguese menu options.

Features

- Test certificate included for -X0, -X1
- Patented Magmeter technology*
- No moving parts
- Bi-directional flow
- Empty pipe detection
- Installs into pipe sizes DN15 to DN900 (0.5 to 36 in.)
- Operating range 0.05 to 10 m/s (0.15 to 33 ft/s)
- Accurate measurement even in dirty liquids
- Polypropylene or PVDF retaining nuts
- Blind 4 to 20 mA, digital (S³L), frequency, relay output
- No pressure drop
- Corrosion resistant materials; PP or PVDF with SS, Hastelloy-C, or Titanium
- Multi-language display menu available



Applications

- Chemical Processing
- Water and Wastewater Monitoring
- Metal Recovery and Landfill Leachate
- Commercial Pools, Spas, and Aquariums
- HVAC
- Irrigation
- Scrubber Control
- Neutralization Systems
- Industrial Water Distribution

* U.S. Patent No: 7,055,396 B1

Specifications

General		
Operating Range	0.05 to 10 m/s	0.15 to 33 ft/s
Pipe Size Range	DN15 to DN900	½ in. to 36 in.
Linearity	± 1% reading plus 0.1% of full scale	
Repeatability	±0.5% of reading @ 25 °C [77 °F]	
Minimum Conductivity	20 µS/cm	
Wetted Materials		
Sensor Body/Electrodes and Grounding Ring	-P0, -P1, -P2: PP/316L SS -T0, -T1, -T2: PVDF/Titanium -V0, -V1, -V2: PVDF/Hastelloy-C	
O-rings	FPM (standard) EPR (EPDM), FFPM (optional)	
Case	PBT	
Display Window	Polyamide (transparent nylon)	
Protection Rating	NEMA 4X/IP65	
Electrical		
Power Requirements	4 to 20 mA	24 VDC ±10%, regulated, 22.1 mA max.
	Frequency	5 to 24 VDC ±10%, regulated, 15 mA max.
	Digital [S ³ L]	5 to 6.5 VDC, 15 mA max.
Auxiliary (only required for units with relays)		9 to 24 VDC, 0.4 A max.
Reverse Polarity and Short Circuit Protected		
Current Output 4 to 20 mA	Loop Accuracy	32 µA max. error [25 °C @ 24 VDC]
	Isolation	Low voltage < 48 VAC/DC from electrodes and auxiliary power
	Maximum Cable	300 m [1000 ft]
	Error condition	22.1 mA
	Max. Loop Resistance	300 Ω
	Compatible with PLC, PC or similar equipment	
	4 to 20 mA load needed	
Frequency Output	Output Modes	Freq., or Mirror Relay (display version only)
	Max. Pull-up Voltage	30 VDC
	Max. Current Sink	50 mA, current limited
	Maximum Cable	300 m [1000 ft]
	Compatible with Signet Model 8550, 8900, 9900, 9900-1BC	
Digital [S ³ L] Output	Serial ASCII, TTL level 9600 bps	
	Compatible with Model Signet 8900 controller	
Relay Specifications		
#1, #2 Type	Mechanical SPDT	
Rating	5 A @ 30 VDC max., 5 A @ 250 VDC max.	
#3 Type	Solid State	
	50 mA @ 30 VDC, 50 mA @ 42 VAC	
Hysteresis	User adjustable for exiting alarm condition	
Alarm On Trigger Delay	Adjustable [0 to 9999.9 sec.]	
Relay Modes	Off, Low, High, Window, and Proportional Pulse	
Relay Source	Flow Rate, Resettable Totalizer	
Error Condition	Selectable; Fail Open or Closed	
Display		
Characters	2 x 16	
Contrast	User-set in four levels	
Backlighting (only on relay versions)	Requires external 9-24 VDC, 0.4 mA max.	
Max. Temperature/Pressure Rating		
Storage Temperature	-20 °C to 70 °C	-4 °F to 158 °F
Relative Humidity	0 to 95% [non-condensing]	
Operating Temperature	Ambient	-10 °C to 70 °C
	Media	0 °C to 85 °C
Maximum Operating Pressure	10.3 bar @ 25 °C	150 psi @ 77 °F
	1.4 bar @ 85 °C	20 psi @ 185 °F
Shipping Weight		
	0.680 kg	1.50 lb
Standards and Approvals		
CE, FCC, UL, CUL [for display versions with relays]		
RoHS compliant, China RoHS		
NEMA 4X / IP65 Enclosure [with cap installed]		
Manufactured under ISO 9001 for Quality and ISO 14001 for Environmental Management and OHSAS 18001 for Occupational Health and Safety		